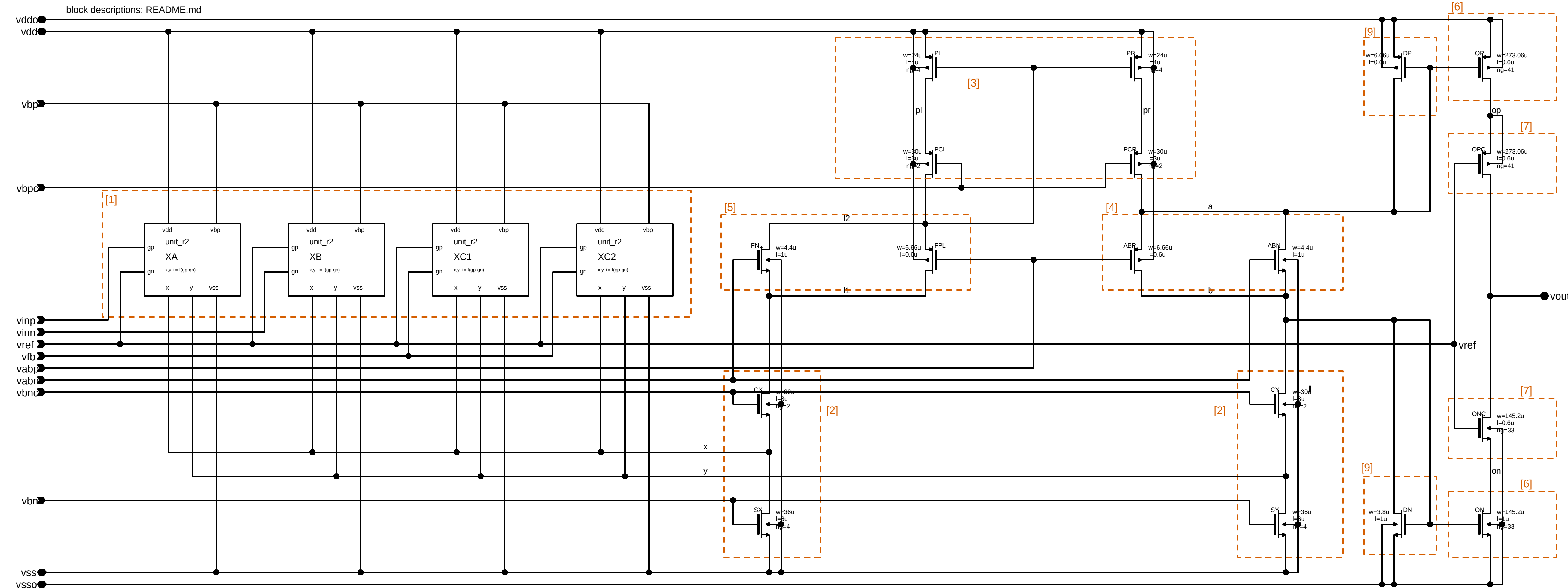


d2s\_lc2: case (b), cascoded output devices in the slot (half frames), cascode gates at vref, load-compensated (no Miller capacitor)

frozen .subckt defaults: lcas = 3 (CX, CY, PCL, PCR 30u / 3u ng=2), wdp = 6.66u, wdn = 3.8u; units are unit\_r2 (rl = 50.6u)

block descriptions: README.md



- [1] DDA units A, B, C1, C2: split-tail PMOS pair, rhigh between the sources. Each adds  $f(gp - gn)$  to the fold nodes: the gp device drains to y, the gn device to x.  
A:  $v_{inp} - v_{ref}$ , B:  $v_{ref} - v_{inn}$ , C1 = C2:  $v_{ref} - v_{fb}$ . The loop forces  $f(v_{inp} - v_{ref}) + f(v_{ref} - v_{inn}) = 2 f(v_{fb} - v_{ref})$ , so  $v_{fb} - v_{ref} = (v_{inp} - v_{inn})/2$ .
- [2] fold: sinks SX, SY (gate vbn) and NMOS cascodes CX, CY (gate vbnc). x and y are the fold nodes.
- [3] PMOS cascoded mirror: PL, PCL is the input side (gates tied to I2), PR, PCR the output side into node a. Cascode gates vbpc.
- [4] class-AB control: ABP (gate vabp) and ABN (gate vabn) between a and b. With RP1/RP2 and RN1/RN2 in the bias it sets the quiescent current of OP and ON.
- [5] FPL, FNL: copy of [4] in the mirror input branch (between I2 and I1), so both fold branches carry the same element.
- [6] output devices OP (gate a), ON (gate b): half frames, in the slot, not the pad clamps.
- [7] output cascodes OPC, ONC, gates at vref (output range about 1.0 ... 2.4 V).
- [9] diode replicas DP (on a), DN (on b), 1/41 and 1/38 of OP, ON: low-impedance gate nodes. No compensation capacitor: the load capacitance is the dominant pole.